Introduction to Computer and Computer Science

Homework Assignment #5

Due: 10/28 0:00 am

In this homework assignment, you will practice how to apply the concept of recursion to solve some practical issues. As you have learned all the related knowledge during the lectures (exhaustive enumeration, bisection search and so on…), you should have the ability to accomplish this.

Do we have common biological features?

DNA, a gigantic molecule aims to store the genetic information, commonly consists of a sequence of one of four nucleotides - adenine (A), cytosine (C), guanine (G), or thymine (T). For simplicity, a DNA strand is represented by a string composed of elements from an alphabet of only four symbols, for example, the string AAACAACTG represents a particular strand of DNA. One way to understand the function of a particular strand of DNA (or a sub-strand of DNA) is to match that strand against a library of known DNA sequences, i.e., a library consists of billions of DNA sequences, whose function and structure are well-known. However, searching a matching DNA sequence through a huge database is not an easy work, for example, the combinations of human chromosome is believed to have on the order of several hundred million. Therefore, an efficient algorithm for searching the matching string will be very useful and helpful.

In this homework, you will be asked to write two functions for accomplishing this string-matching search task iteratively and recursively.

Problem #1: Iterative search for matching string

Please write a function called “**searchSSM**” that aims to return you the indices of the starting location of matching string. It will take two arguments, a key string and a target string, as shown below:

searchSSM(**target**, **key**)

Problem #2: Recursive search for matching string

Please write a function called “**searchSSMR**” that aims to return you the indices of the starting location of matching string. It will take two arguments, a key string and a target string, as shown below:

searchSSMR(**target**, **key**)

You can test your function with these cases:

Case 1

Target: ‘1234567890123456789’

Key: ‘5’

And your function should return you: (4, 14)

Case 2

Target: ‘atgacatgcacaagtatgcat’

Key: ‘atg’

And your function should return you: (0, 5, 15)

Case 3

Target: ‘atgaatgcatggatgtaaatgcag’

Key: ‘tg’

And your function should return you: (1, 5, 9, 13, 19)

Hint:

You can add any default parameter into these two functions.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
# ==========================================================================
#
# Homework assignment #0
# Student: Your name, 你的名字
# Student ID: Your student ID, 你的學號
# Collaborators: John Doe, Jane Doe, ...
#
# ==========================================================================
# Your code goes here...

```

Please save your code as a ".py" file, where the file name should follow this format:

108A_hw#?_YourID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature**. Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the "New e3" system. There will be a section for uploading your homework.